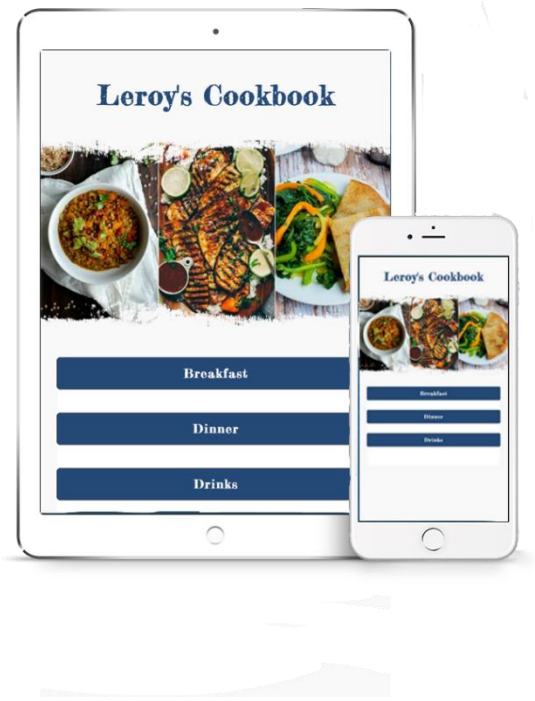
# Development Plan

## Leroy’s Cookbook

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Website url: http://localhost/server/cookbook/index.html

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# Introduction

This report details the design, implementation and testing of the mobile application developed for Leroy's Cookbook.

Leroy's Cookbook has been developed to provide recipes for meals to cook for breakfast and dinner. Each meal will have an image, a list of ingredients and the steps required.

# System Requirements

The following functional and non-functional requirements will used for the system.

## Functional requirements

1. Navigation and interactive elements must produce consistent results.
2. System must display a list of available meal categories.
3. System must display a list of meals for each category.
4. System must allow user to search list of meals.
5. System must display search results from list.
6. System must allow user to view recipe for each meal in list.

## Non-functional requirements

1. User interfaces must be aesthetically consistent.
2. System must conform to W3C standards.
3. System must be accessible and useable.
4. System must be fully responsive for desktop and mobile devices.
5. System must provide support for user-agents.

# Prototyping

Prototyping demonstrates the functionality of the system.

## Site Map

The site map shown in figure 1 represents the structure of the website.

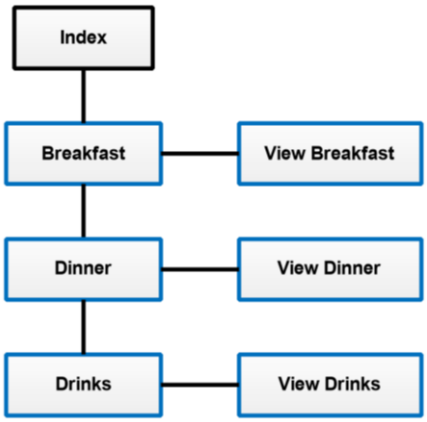


Figure 1. The site map for the application.

## User Interfaces

The graphical user interface provides the necessary textual and visual information to assist in achieving the required functionality.

### Landing/Index Page

The landing page must display the main title and the available meal categories.

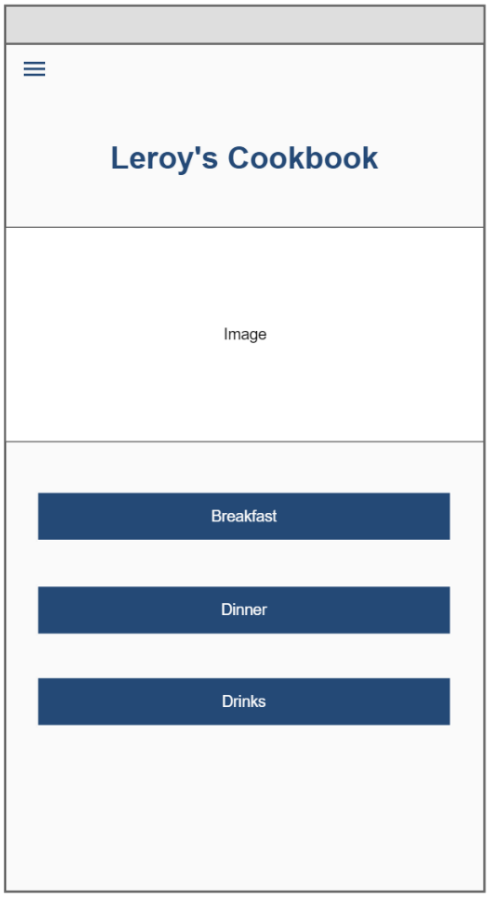


Figure 2. The Landing page for the application.

### List Page

The List Page must display a list of available meals within the selected category.

Figure 3. A list of available meals.

### View Page

The View Page must display an image, the ingredients and steps required for a selected meal.

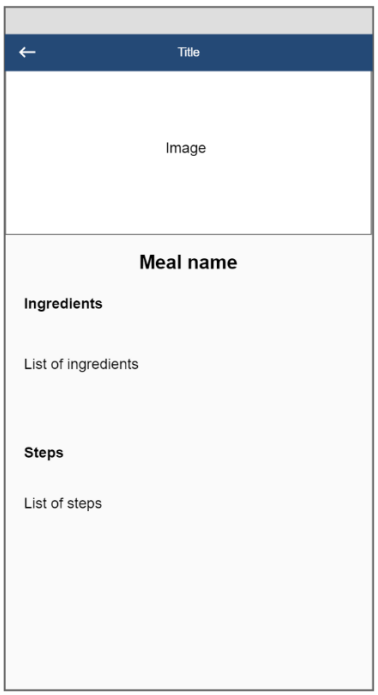


Figure 4. Recipe for a selected meal.

# Implementation

The following technologies will be used to build the structure, presentation and functionality of the system.

## Front-end

1. HTML5
2. CSS3
3. JavaScript
4. jQuery Mobile
5. JSON
6. Krita

| Technology | Usage |
|---|---|
| HTML5 | Structure and layout of the web pages |
| CSS3 | Styling and responsiveness of the web pages |
| JavaScript | Functionality for the website |
| jQuery Mobile | Framework used to develop mobile app |
| JSON | Structure for data |
| Krita | Computer software used for images manipulation and creation |

Table 1. Technologies used for the front-end development.

### Back-end

1. JavaScript
2. JSON

| Technology | Usage |
|---|---|
| JavaScript | Retrieval of the data from JSON |
| JSON | Structure of the data to be displayed |

Table 2. Technologies used for the back-end development.

## Testing

The following tests ensure the website meets web standards, browser compatibility and accessibility.

### Compliance

HTML5 and CSS3 have been validated and comply with W3C standards.

### Accessibility

ARIA landmark roles, ARIA properties and skip links have been implemented to enhance website accessibility and usability.

### Browser Compatibility

All web browsers are tested for consistency and responsiveness. The test results are shown in Table 3.

| Web Browser | Version | Result |
|---|---|---|
| Firefox | 79 | Pass |
| Chrome | 84 | Pass |
| Microsoft Edge | 44 | Pass |
| Internet Explorer | 11 | Pass |
| Opera | 69 | Pass |
| Safari | 5.3 | Fail<br>Issue: Alignment and displaying of elements. |

Table 3. The test results for web browser compatibility.

## Enhancements

Further improvements for the website are listed in table 4.

| Task | Improvement |
|---|---|
| Data Storage | • Use of a database to store and retrieve data for recipes. |
| Browser Compatibility | • HTML5 shiv or other techniques should be used to make website compatible with older browsers. |
| Functionality | • Include ways to search for meal to cook.<br>• Include video demonstration for recipes. |

Table 4. List of recommended enhancements.